Practice LQ2 – Form 2 CS 210 / Math 252 – Discrete Mathematics

Comprehensive Practice: Lectures 13–20 Fall 2026

Student Name: _____ **Student ID:** _____

25 questions × 4 points = 100 points. Work every problem as a timed practice set. Ten prompts are identical to Long Quiz 2; the remaining prompts are parallel Rosen-inspired practice. Show definitions, witnesses, and intermediate work.

1. On $A = \{1, 2, 3, 6\}$ define xRy when $x \mid y$. Decide whether R is reflexive, symmetric, antisymmetric, and transitive. Justify each answer and give a specific witness for each failed property.

2. On $\mathbb{Z}$, define xRy when $x - y$ is even. Prove that R is an equivalence relation and describe $[7]$.

3. On the set of all finite bit strings, define $s \sim t$ when the numbers of 1s in s and t are congruent modulo 3. Prove that $\sim$ is an equivalence relation and describe the class containing 10110.

4. On $\{1, 2, 3, 4\}$ start with $R = \{(1, 2), (2, 3), (3, 4)\}$. List every pair added to form the smallest relation containing R that is both reflexive and transitive.

5. Order $S = \{1, 2, 3, 4, 6, 12\}$ by divisibility. List every cover pair $x \prec y$ (this specifies the Hasse diagram), then give one comparable pair and one incomparable pair.

6. In the divisibility poset on $S = \{1, 2, 3, 6, 12, 18\}$, find the minimal and maximal elements, decide whether least/greatest elements exist, and find all upper bounds and the least upper bound of $\{2, 3\}$.

7. Let $f : \mathbb{Z} \to \mathbb{Z}$ be $f(n) = 3n - 2$. Prove or disprove injectivity and surjectivity, and describe the range. Change only the codomain so that the same rule is bijective, then give its inverse on that codomain.

8. Let $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ be $f(m, n) = m - n$. Determine whether f is injective and surjective, with a proof or witness for each answer.

9. For $f, g : \mathbb{R} \to \mathbb{R}$ given by $f(x) = 2x - 1$ and $g(x) = x^2 + 3$, find $f \circ g$ and $g \circ f$. Give f^{-1}. Explain why g has no inverse as a function $\mathbb{R} \to \mathbb{R}$.

10. Show that $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^3 + 2$, is bijective and find its inverse.

11. Reindex $\displaystyle\sum_{k=3}^{n}(2k+1)$ using $j = k - 3$. Give the new summand and limits, then evaluate the sum in closed form for $n \geq 3$.

12. Let $a_0 = 1$ and $a_n = 3a_{n-1}$. List a_0 through a_4 and give a closed form.

13. Evaluate $\displaystyle\sum_{k=1}^{n}\frac{1}{(k+1)(k+2)}$ by partial fractions and telescoping. Give a closed form and its value at $n = 8$.

14. Evaluate $\displaystyle\sum_{i=1}^{n}\sum_{j=1}^{i}1$ in closed form and explain the count it represents.

15. A robot crosses a strip of n cells using jumps of exactly 1 or 3 cells. Let w_n be the number of jump sequences that land exactly at the end. Give a recurrence with sufficient initial conditions and compute w_7.

16. A machine accepts tokens worth 1 or 2 units, one at a time. Let d_n count ordered token sequences totaling n. Give a recurrence and initial conditions, then find d_6.

17. Let $U(1) = 2$ and $U(n) = 2U(n/2) + n$ for powers of two. Using a recurrence tree, give the work at each internal level, the total leaf cost, an exact formula, and a tight Θ bound.

18. Solve $T(1) = 1$, $T(n) = 3T(n/3) + 2n$ for $n = 3^k$ using a recurrence tree. Give an exact formula and tight bound.

19. A binary-search implementation uses at most $C(n) = \lceil \log_2 n \rceil$ halving rounds on n sorted records. Find $C(1{,}000{,}000)$ and $C(2{,}000{,}000)$, state the asymptotic class, and explain the practical effect of doubling the database.

20. An algorithm compares each unordered pair of distinct items once. Find the exact number of comparisons for n items and a tight bound.

21. Let $f : \mathbb{Z} \to \{0, 1, 2, 3\}$ be $f(n) = n \bmod 4$, and define $x \sim y$ when $f(x) = f(y)$. Prove equivalence and describe every class.

22. On residues modulo 8, let $h(n) = 3n + 4$. Prove h is bijective, find its inverse, and decode the output 1.

23. Five tasks satisfy $A < C$, $B < D$, $C < E$, and $D < E$. Count all valid schedules and justify the count structurally.

24. Let $T(1) = 1$ and $T(n) = 4T(n/2) + n^2$ for $n = 2^k$. Find an exact formula from a recurrence tree and give a tight bound.

25. Let R_{12} and R_4 be congruence modulo 12 and modulo 4 on $\mathbb{Z}$. Prove that the partition for R_{12} refines the partition for R_4, and state which three R_{12} classes form $[1]_4$.